

## **Animal Physiology and Biochemistry**

### **1. Introduction**

Animal physiology is the study of how animal body systems function and work together to maintain life. Biochemistry is the study of the chemical substances and reactions that occur in living organisms, including animals.

Together, these subjects explain how animals obtain energy, digest food, respire, circulate substances, remove wastes, regulate temperature, coordinate body functions, and use hormones to control growth and reproduction. They are central to zoology because they connect structure, function, and metabolism.

### **2. Scope and importance**

Animal physiology and biochemistry are important for understanding normal body function as well as disease, adaptation, and survival. They are also essential in medicine, veterinary science, nutrition, biotechnology, wildlife biology, and research.

These subjects explain how enzymes work, how nutrients are metabolized, how oxygen is transported, how nerves send signals, and how hormones regulate body processes.

### **3. Basic biochemical concepts**

Biochemistry deals with biomolecules such as carbohydrates, lipids, proteins, nucleic acids, enzymes, vitamins, and minerals. In animal systems, these molecules build tissues, store energy, catalyze reactions, carry messages, and maintain homeostasis.

#### **Major biomolecules**

- Carbohydrates: quick energy source.
- Lipids: long-term energy storage and membranes.
- Proteins: structure, enzymes, transport, hormones.
- Nucleic acids: genetic information.
- Vitamins: metabolic regulators.
- Minerals: structural and catalytic roles.

### 4. Enzymes

Enzymes are biological catalysts that increase the rate of biochemical reactions without being consumed. Most enzymes are proteins, though some RNA molecules also act as catalysts.

#### Characteristics of enzymes

- Highly specific.
- Work best at an optimum pH and temperature.
- Lower activation energy.
- Can be regulated.
- Often require cofactors or coenzymes.

#### Importance in animals

Digestive enzymes break down food, metabolic enzymes support energy production, and regulatory enzymes control biochemical pathways.

### 5. Carbohydrates, lipids, and proteins

These macromolecules are central to animal metabolism. Carbohydrates are broken down to glucose, lipids to fatty acids and glycerol, and proteins to amino acids.

#### Carbohydrates

They provide immediate energy and are stored as glycogen in animals.

#### Lipids

They are major energy reserves and are important in membranes and hormone synthesis.

#### Proteins

They build muscle, enzymes, transporters, and hormones, and they also serve as a backup energy source when needed.

### 6. Digestion

Digestion is the process of breaking complex food into simple, absorbable molecules. Animal digestion varies across species, but the general purpose is the same: to make nutrients available for metabolism and growth.

#### Stages of digestion

1. Ingestion.
2. Mechanical digestion.
3. Chemical digestion.
4. Absorption.

5. Assimilation.
6. Egestion.

### **Digestive enzymes**

- Amylase digests starch.
- Proteases digest proteins.
- Lipases digest fats.
- Nucleases digest nucleic acids.

### **Important idea**

Digestion is not only about breaking food apart; it also prepares nutrients for transport across membranes and conversion into energy or body tissue.

### **7. Absorption and assimilation**

Absorption is the movement of digested nutrients from the gut into the blood or lymph. Assimilation is the use of absorbed nutrients by body cells.

### **8. Respiration**

Respiration is the process by which animals obtain oxygen, release carbon dioxide, and produce energy in the form of ATP. It includes external respiration, transport of gases, and cellular respiration.

#### **Types of respiration**

- Cutaneous respiration.
- Gills.
- Tracheal system.
- Lungs.
- Book lungs.

#### **Gas transport**

Oxygen and carbon dioxide are transported in body fluids, often with the help of respiratory pigments such as hemoglobin, hemocyanin, or hemerythrin.

## **9. Circulation**

Circulation is the movement of blood or body fluid that transports oxygen, nutrients, hormones, and wastes. It is essential for maintaining homeostasis.

### **Heart types**

- Neurogenic heart.
- Myogenic heart.

### **Cardiac cycle**

The cardiac cycle includes contraction and relaxation of the heart chambers, ensuring continuous circulation of blood.

### **Homeostasis**

The circulatory system helps maintain temperature, pH, fluid balance, and nutrient distribution.

## **10. Excretion and osmoregulation**

Excretion removes metabolic wastes from the body. Osmoregulation maintains the balance of water and salts in body fluids.

### **Nitrogenous wastes**

- Ammonia.
- Urea.
- Uric acid.

### **Kidney function**

The kidney filters blood, removes wastes, conserves useful substances, and forms urine.

### **Urea cycle**

Urea is formed in the liver through the urea cycle and then excreted by the kidneys.

## **11. Nervous system**

The nervous system coordinates body activities through electrical impulses and chemical signals. It allows animals to sense the environment and respond quickly.

### **Neuron structure**

A neuron has a cell body, dendrites, and axon. It transmits nerve impulses from one part of the body to another.

### **Nerve impulse**

A nerve impulse is an electrochemical signal that travels along the neuron and across synapses.

### **Sensory receptors**

Receptors detect stimuli such as light, sound, pressure, chemicals, and temperature.

### **12. Muscle physiology**

Muscles generate movement and force. Animal muscle tissues include skeletal, smooth, and cardiac muscles.

#### **Muscle features**

- Excitability.
- Contractility.
- Extensibility.
- Elasticity.

#### **Sliding filament mechanism**

Muscle contraction occurs when actin and myosin filaments slide over each other using energy from ATP.

### **13. Thermoregulation**

Thermoregulation is the ability of an animal to maintain a suitable body temperature. Animals may be poikilothermic or homeothermic depending on how they regulate temperature.

#### **Key terms**

- Poikilotherms: body temperature varies with environment.
- Homeotherms: maintain nearly constant body temperature.
- Cold resistance.
- Heat resistance.
- Cold death.
- Heat death.

#### **Importance**

Thermoregulation supports survival in different habitats and is linked to metabolism and enzyme function.

## **14. Endocrinology**

Endocrinology is the study of hormones and endocrine glands. Hormones are chemical messengers that regulate growth, metabolism, reproduction, and homeostasis.

### **Major endocrine glands**

- Pituitary.
- Thyroid.
- Parathyroid.
- Adrenal.
- Pancreas.
- Gonads.

### **Hormone types**

- Peptide hormones.
- Steroid hormones.
- Amino acid-derived hormones.

## **15. Pituitary and thyroid glands**

The pituitary gland is often called the master gland because it regulates many other endocrine glands. The thyroid gland controls metabolism, growth, and development.

### **Pituitary hormones**

- Growth hormone.
- TSH.
- ACTH.
- FSH.
- LH.
- Prolactin.

### **Thyroid hormones**

- Thyroxine.
- Triiodothyronine.
- Calcitonin.

## **16. Pancreas and adrenal glands**

The pancreas has both endocrine and exocrine functions. The adrenal glands produce hormones involved in stress response, salt balance, and metabolism.

### **Pancreatic hormones**

- Insulin.
- Glucagon.

### **Adrenal hormones**

- Cortisol.
- Aldosterone.
- Adrenaline.
- Noradrenaline.

### **Importance**

These hormones regulate blood glucose, stress adaptation, and electrolyte balance.

## **17. Reproductive endocrinology**

Reproductive endocrinology studies hormones controlling reproduction, including testicular hormones, ovarian hormones, pregnancy, parturition, lactation, and pheromones.

### **Main topics**

- Structure of testes and ovary.
- Male and female accessory organs.
- Estrous and menstrual cycles.
- Hormones of pregnancy.
- Parturition.
- Lactation.
- Pheromones.

## **18. Pheromones**

Pheromones are chemical signals released by one individual that affect the behavior or physiology of another individual of the same species.

### **Significance**

They are important in reproduction, communication, territorial behavior, and social interactions.

## **19. Interrelationship between physiology and biochemistry**

Physiology and biochemistry are deeply connected. Biochemistry explains the molecules and reactions that make physiological functions possible, while physiology explains how those reactions work at the level of organs and whole organisms.

For example, enzymes drive digestion, ATP supports muscle contraction, hemoglobin helps respiration, and hormones regulate metabolism.

## **20. Applications**

Animal physiology and biochemistry are used in:

- Medicine.
- Veterinary science.
- Nutrition.
- Wildlife biology.
- Biotechnology.
- Environmental science.
- Research laboratories.
- Pharmacology.
- Conservation studies.

## **21. Summary**

Animal Physiology and Biochemistry explain how the animal body works at both the molecular and organ-system levels. The subject covers biomolecules, enzymes, digestion, respiration, circulation, excretion, nerves, muscles, thermoregulation, and endocrine control. Together, these topics form one of the most important foundations of zoology and animal science.